Huawei Optical Transport Failure Resolution for Access Node JAY-ACC-39

Node: JAY-ACC-39 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_7328

Description

The failure domain of Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on the Huawei Access Node JAY-ACC-39, situated at Jayanagar Metro Site, is a critical severity alarm (ALARM_7328) that necessitates immediate attention to prevent protracted network downtime and data loss. This failure domain is often precipitated by a concatenation of factors including, but not limited to, the degradation of optical signal quality due to aging or damaged SFP/XFP modules, physical obstructions or cuts in the fiber optic cables, and power loss affecting the reception and transmission of optical signals. The Huawei equipment, leveraging its proprietary CLI dialect, provides a robust set of diagnostic and resolution commands to address these issues. The optical transport layer, critical for the backbone of modern telecommunications networks, relies on the precise calibration and maintenance of its components to ensure high-speed data transmission without signal degradation. Therefore, understanding the intricacies of the Optical Transport Failure, including the mechanics of signal propagation, amplification, and the potential points of failure within the Huawei Access Node, is essential for the effective troubleshooting and resolution of this critical alarm.

Pre-requisite Checks

- Dispatch remote hands to Jayanagar Metro Site for on-site inspection and intervention.
- Verify the status of redundant links to ensure network resilience.
- Check the environmental conditions (temperature, humidity) at the site.
- Confirm the availability of spare SFP/XFP modules and fiber optic cables.

Potential Root Causes

Cause	Probability
SFP/XFP Module Degradation	High
Fiber Optic Cable Damage	Medium
Rx/Tx Power Loss	Low

Diagnostic Commands & Expected Outputs

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Command: display interface brief, Expected Output: List of interfaces with their current status, Description: This command is utilized to quickly assess the status of all interfaces on the device, helping to isolate any that are down or experiencing errors, which could indicate the source of the Optical Transport Failure.
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Command: display optical-interfaces, Expected Output: Detailed information about the optical interfaces, Description: This command provides detailed diagnostics about the optical interfaces, including signal strength, which is crucial for identifying issues related to SFP/XFP degradation or fiber cuts.
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Resolution Steps

Step 1: Step: Replace Degraded SFP/XFP Module, Command: replace sfp slot 1/0/1, Description: If the diagnostics indicate a degraded SFP/XFP module, replacing it with a new one can resolve the issue. The command to replace the module must be executed with caution, ensuring the new module is compatible and properly seated.

Step 2: Step: Repair or Replace Damaged Fiber Optic Cable, Command: N/A, Description: For issues related to fiber optic cable damage, physical intervention is required. This involves either repairing the damaged section of the cable or replacing it entirely, ensuring that the new or repaired cable is properly connected and tested.

Post-Check Verification

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Command: display interface brief, Expected Output: Verification that the previously faulty interface is now up, Description: After executing the resolution steps, this command is used again to verify that the interface status has returned to normal, indicating a successful resolution of the Optical Transport Failure.

Command: display optical-interfaces, Expected Output: Improved or normalized optical signal quality, Description: This command is utilized post-resolution to confirm that the optical signal quality has improved or returned to normal parameters, ensuring that the fix has been effective.
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Roll-back Procedures

- Step: Revert to Original SFP/XFP Module, Command: replace sfp slot 1/0/1 original-module, Description: If the replacement of the SFP/XFP module does not resolve the issue or introduces new problems, reverting to the original module may be necessary. This involves removing the newly installed module and reinstalling the original one.

- Step: Restore Original Fiber Optic Cable Configuration, Command: N/A, Description: In cases where the fiber optic cable was replaced or repaired, and the issue persists or worsens, restoring the original cable configuration may be required. This involves carefully removing any new cable sections and reconnecting the original cables, ensuring all connections are secure and tested.